

■ 2.6.2 Short Output Form

When you generate a very large output expression in *Mathematica*, you often do not want to see the whole expression at once. Rather, you would first like to get an idea of the general structure of the expression, and then, perhaps, go in and look at particular parts in more detail.

The function **Short** gives you a way to have *Mathematica* print shortened forms of expressions.

<code>Short[expr]</code>	print a one-line outline of <i>expr</i>
<code>Short[expr, n]</code>	print an <i>n</i> -line outline of <i>expr</i>

Showing expressions in short form.

This generates a long expression. If the whole expression were printed out here, it would go on for 23 lines.

```
In[1]:= t = Expand[(1 + x + y)^12] ;
```

This gives a one-line “outline” of *t*. The <<86>> indicates that 86 terms are omitted.

```
In[2]:= Short[t]
Out[2]//Short=
1 + 12 x + 66 x^2 + <<86>> + 12 x y^11 + y^12
```

Here is a four-line version of *t*. More terms are shown in this case.

```
In[3]:= Short[t, 4]
Out[3]//Short=
1 + 12 x + 66 x^2 + 220 x^3 + 495 x^4 + 792 x^5 +
924 x^6 + 792 x^7 + 495 x^8 + <<77>> + 132 x y^10 +
66 x^2 y^10 + 12 y^11 + 12 x y^11 + y^12
```

You can use **Short** with other output forms, such as **InputForm**.

```
In[4]:= Short[InputForm[t]]
Out[4]//Short= 1 + 12*x + 66*x^2 + <<87>> + y^12
```

When *Mathematica* generates output, it first effectively writes the output in one long row. Then it looks at the width of text you have asked for, and it chops the row of output into a sequence of separate “lines”. Each of the “lines” may of course contain superscripts and built-up fractions, and so may take up more than one actual line on your output device. When you specify a particular number of lines in **Short**, *Mathematica* takes this to be the number of “logical lines” that you

want, not the number of actual physical lines on your particular output device.